

Homigo Care App Blueprint

Android Native App · iOS Native App · Responsive Web App — Build Specification for AI Coding Agents

Prepared from 13 source documents supplied by the Founder · Consolidated Edition

2026

- [0. How This Document Was Built](#)
- [1. Product Recap \(Source of Truth — Unchanged\)](#)
- [2. Complete User & Role Architecture](#)
 - [2.1 Patients — 3 Registration Categories](#)
 - [2.2 Professional Roles \(Dynamic Role Engine\)](#)
 - [2.3 The 3-Phase Professional Assessment \(mandatory, unchanged from source\)](#)
- [3. Complete Feature Set \(Consolidated\)](#)
 - [3.1 Accessibility \(core differentiator — build these first-class, not as an afterthought\)](#)
 - [3.2 Booking Engine](#)
 - [3.3 Health Passport \(flagship feature\)](#)
 - [3.4 Safety System \(female healthcare worker protection — Pakistan-specific\)](#)
 - [3.5 Marketplace Verticals](#)
 - [3.6 Pricing & Commission \(unchanged — do not alter without founder sign-off\)](#)
 - [3.7 Payments, Wallet & Payouts](#)
 - [3.8 Ratings, Reviews & Notifications](#)
 - [3.9 Referral System](#)
 - [3.10 Settings Decisions \(carried over as-is\)](#)
- [4. Platform Strategy — Three Separate Builds](#)
 - [4.1 Android Native App](#)
 - [4.2 iOS Native App \(Premium Tier\)](#)
 - [4.3 Web App — One Codebase, Three Responsive Personalities](#)
- [5. Design System \(unchanged brand tokens — apply identically on all 3 platforms\)](#)
 - [5.1 Mobile-Safe Layout Rules \(mandatory on every screen, every platform\)](#)
- [6. Icons & Visual Assets Strategy](#)
 - [6.1 Icon Libraries — Use the Real, Official, Free Ones](#)
 - [6.2 Illustrations & PNG Assets — Two Legitimate Paths](#)
 - [6.3 Batched AI Image-Generation Prompt System](#)
- [7. Backend Architecture — 100% Free-Tier Stack](#)
 - [7.1 Division of Responsibility](#)
 - [7.2 What Stays “Optional / Plug In Later” \(per your instruction\)](#)
 - [7.3 Payments — Manual System \(as instructed\)](#)
 - [7.4 Location & Maps — Free Tier](#)
 - [7.5 Auto-Created Collections & Tables \(adapted from the original DB schema\)](#)
- [8. Security \(adapted to the free-tier stack\)](#)
- [9. Zero Fake Data — Real-Base System Policy](#)
- [10. Screen Inventory Reference](#)
 - [10.1 Patient App/Site — 32 Screens](#)
 - [10.2 Professional App/Site — 15 Screens](#)
 - [10.3 Admin Panel — see **Document 2** for the full, dedicated specification.](#)
- [11. Recommended AI Coding Assistant & Model](#)
- [12. Build Roadmap \(phases carried over from the original spec, re-mapped to the new stack\)](#)

0. How This Document Was Built

This blueprint was produced after reading **every file** supplied in guide (1).zip:

#	File	Content
1	HomigoCare_User_Definitions.docx	16 user types, documents required, 3-phase assessment
2	HomigoCare_Complete_Feature_Discussion-1.docx	Founder’s full reasoning behind every feature (13 parts)
3	Homigo_Care_Core_Features(1).pdf	Short feature checklist
4	HomigoCare_Developer_Guide.docx	Original tech stack, design system, 46 screen specs, DB schema, APIs
5	HomigoCare_Complete_Developer_Document.docx	Screen-by-screen layout for all 32 patient + 15 professional + 5+ admin screens, user flows, legal summary
6	HomigoCare_Registration_Guide.pdf	3 patient types + 20+ professional roles, dynamic registration engine
7	HomigoCare_HMS_QR_Feature_Urdu.pdf	Hospital Management System + universal QR patient record (Phase 2)
8	HomigoCare_Flowcharts_and_Screens.html	Role flowcharts, realistic screen-by-screen walkthroughs
9	HOMIGO PROMPTS.docx	57 ready-made AI image prompts (per-screen mockups) — style reference
10	CODING.docx / CODING 2.docx (identical)	Interactive HTML prototype/simulator source code

11	homigo_care_interactive_clickable_prototype(1).html	Clickable prototype, same content as above
12	Screens .pdf	30 pages of actual rendered mockup screenshots (visual ground truth)

Total unique pages/content reviewed: ~47 PDF pages + 7 Word documents (~1,900 paragraphs/rows) + 2 HTML documents + 30 visual mockup screens.

Important note on conflicts found: The original Developer Guide and Complete Developer Document recommend **React Native + Node.js + AWS + PostgreSQL + Twilio**. Your new instructions (this conversation) explicitly ask for **two separate native official apps (Android + iOS) instead of one cross-platform app**, and a **100% free-tier stack (Firebase, Supabase, Vercel, Cloudinary, EmailJS)** instead of AWS. This document **overrides the old tech stack** with the new one below, while keeping 100% of the product logic, screens, pricing, and rules from the original documents intact.

1. Product Recap (Source of Truth — Unchanged)

Name: Homigo Care — *“Because Care Begins at Home”* **Market:** Islamabad & Rawalpindi, Pakistan (2026), expanding later to Karachi/Lahore. **What it is:** A platform (three separate front-ends, one shared backend) connecting **patients** with **verified home-healthcare professionals** (nurses, doctors, labs, pharmacies, equipment providers) plus a **B2B hospital-supply portal** and a **5-level admin system**.

Two core values everything is built around: 1. **Credibility** — every professional is document-verified AND must pass a 3-phase test before going live. 2. **Professionalism** — consistent design system, mandatory visit notes, GPS safety, 5-category ratings.

Three ecosystem layers: - Layer 1 — B2C healthcare (patients booking nurses/doctors/labs/medicine/equipment) - Layer 2 — Healthcare professional platform (verified workers, earnings, reputation) - Layer 3 — B2B healthcare supply (hospitals/clinics ordering supply packs)

Zero-fake-data rule (new instruction, applies everywhere in this document): The system ships with **no seed data, no demo users, no placeholder bookings/reviews**. Every list, dashboard, and chart must render a genuine **empty state** until an admin or a real user creates the first record. This rule applies to Android, iOS, Web, and the Admin Panel equally, and is treated as a hard rule in every section below (see §9).

2. Complete User & Role Architecture

Homigo Care has **far more than the “16 user types”** stated in the original User Definitions doc — the Registration Guide expands nursing/allied-health into 20+ granular roles that must be supported by one **dynamic role-config engine**, not hardcoded screens.

2.1 Patients — 3 Registration Categories

Type	Who	ID Required	Login
Adult Patient (18+)	Anyone 18+ with a CNIC	CNIC front+back	Own phone number
Minor Patient (<18)	Child/teen	No CNIC — Guardian’s CNIC used	Guardian’s phone (child has no login)
Elderly / No-CNIC Patient	Elderly without valid CNIC	Family member’s CNIC + optional old/expired ID	Family member’s phone

Golden registration rule (applies to every role): **ask identity → then contact/OTP → then role-specific documents → then optional extras**, never more than 1-2 fields per screen.

2.2 Professional Roles (Dynamic Role Engine)

Build **one** dynamic registration + document-upload engine driven by a `role_requirements` config table — **not** 20 hardcoded screens. Example row:

```
{
  "role_id": "physiotherapist",
  "role_label": "Physiotherapist",
  "category": "Allied Health",
  "requires_cnic": true,
  "required_documents": ["DPT Degree", "PPTC Registration", "Experience Letter"],
  "requires_affiliation": true,
  "requires_assessment": true,
  "min_age": 21
}
```

Full role list to seed into this config table (document requirements are authoritative from the Registration Guide):

Category	Roles
Nursing Family	Home Visit Nurse, Dedicated Nurse, Hospital Companion Nurse, Hospital Navigator, Student Nurse, Intern Nurse, CNA (Certified Nursing Assistant)

Midwife & Allied Health	Community Midwife (CMW), Lady Health Visitor (LHV), Medical Lab Technologist (MLT), Physiotherapist, Occupational Therapist
Doctors	General Physician (GP), Specialist Doctor (Cardiology, Gynae, Pediatrics, Neurology, Orthopedics, etc.)
Extended Ecosystem	Speech Therapist, Respiratory Therapist, Psychologist/Counselor, Nutritionist/Dietitian, Palliative/Hospice Care Nurse, Vaccination Specialist, Home Health Aide (non-medical)
Business Partners	Lab Partner, Pharmacy Partner, Equipment Rental Provider
B2B Client	Hospital / Clinic / Nursing Home / Maternity Home
Internal Admin (not app-registered)	Super Admin, Verification Admin, Support Admin, Finance Admin, Emergency Admin

Every professional, regardless of role, must show a **currently working at [Hospital/Clinic name]** field — this is the platform’s core proof of active practice.

2.3 The 3-Phase Professional Assessment (mandatory, unchanged from source)

Phase	Format	Time	Pass %
1 — Basic Knowledge	30 MCQs (pool of 200+)	45 min	70%
2 — Practical Scenario	20 scenario questions	60 min	75%
3 — Ethics & Homigo Standards	20 questions	30 min	80%

Badges: **Homigo Care Certified** (pass all 3) → **Excellence Certified** (≥85% avg) → **Homigo Care Elite Professional** (≥95% avg). Re-assessment every 12 months, or immediately if rating drops below 3.5★ or a major complaint lands. Anti-cheat: randomized question pool, server-side timer, single active device, copy-paste disabled.

3. Complete Feature Set (Consolidated)

3.1 Accessibility (core differentiator — build these first-class, not as an afterthought)

- Voice search in Urdu & English (speech-to-text → service matcher)
- Visual icon-based booking (tap a picture, no reading required)
- Full Urdu language toggle (entire UI, RTL-safe layout)
- Family account management (one login, multiple profiles/health records)

3.2 Booking Engine

4 booking types — **Instant, Scheduled, Recurring, Online Consultation** — plus a 5th special case, **Emergency** (bypasses queue). **Combined bookings** let a patient add nurse + lab + medicine to one payment/reference. **10-minute acceptance window** per professional, auto-cascades to the next available professional after 3 rejections. Voice/written notes attach to every booking and auto-transcribe.

3.3 Health Passport (flagship feature)

Personal, continuously-growing digital health record with 5 sections (Basic Info, Medical History, Uploaded Documents, Auto-logged Activity, Vitals graphs). Protected by a **patient-controlled PIN** — professionals see nothing until the patient shares the PIN for that visit; access auto-revokes when the booking ends and every access is logged and notified to the patient. Separate **Emergency PIN** exposes only blood group/allergies/medications/emergency contact. **Critical Health Alert** tags (HIV, Hepatitis B, severe allergy, etc.) warn a professional before acceptance without exposing the diagnosis until PIN entry.

Unique **Health ID**: HGC- ISB-2026-XXXX with an auto-generated **QR code**, downloadable/printable ID card. (Phase-2 roadmap: scanning this same QR at a partner hospital’s terminal triggers a **consent request** to the patient’s phone — see §11 Roadmap; it never bypasses patient consent, and every hospital access is logged exactly like an in-app PIN access.)

3.4 Safety System (female healthcare worker protection — Pakistan-specific)

- **GPS Safe Zone**: 200 m radius activated at “Visit Started”; breach >500 m or GPS silence >10 min auto-triggers an alert.
- **SOS button**: 3-second hold / double-tap confirm, reachable within 2 taps from anywhere, and **must work without internet** via SMS fallback.
- On trigger: professional’s emergency contact + trusted person get an SMS with live GPS link; Emergency Admin gets a sounding push (mobile + web); admin calls professional (2 min) → patient (2 min) → 1122/15 with GPS coordinates. Full incident report auto-generated and stored in the Health Passport.
- One-tap Pakistan emergency numbers built in: Rescue 1122, Police 15, Edhi Ambulance 115, Women Safety Helpline 1099, Homigo Care’s own line.

3.5 Marketplace Verticals

- **Nurses** — 4 sub-types (Home Visit / Dedicated 8-12-24h / Hospital Companion / Hospital Navigator)
- **Doctors** — GP (home visits + online) and Specialist (mainly online, optional premium home visit)
- **Labs** — partnership model (Chughtai, IDC, Shifa, Excel...), 10% commission, no markup, results auto-land in Health Passport, “Share with Doctor” one-tap
- **Pharmacy** — partnership model, dual revenue (8% pharmacy commission + patient delivery fee), prescription upload or auto-linked digital prescription
- **Equipment Rental** — wheelchairs, hospital beds, oxygen, nebulizers, etc., per day/week/month, 10% commission
- **B2B Supply Packs** — Basic Surgical (PKR 2,500), C-Section (6,000), Normal Delivery (3,000), Orthopedic Fracture (2,000), ICU Central Line (4,000), Emergency (5,000) — 10% commission, bank transfer primary payment
- **Supply Pre-Request** — nurse requests supply list mid-visit, patient approves cost before purchase, receipt-photo proof, commission never charged on supply cost

3.6 Pricing & Commission (unchanged — do not alter without founder sign-off)

Service	Budget	Premium
Nurse Home Visit	PKR 800	PKR 1,500
Dedicated Nurse 8h / 12h / 24h	2,500 / 3,500 / 6,000	4,500 / 6,000 / 10,000
Hospital Companion (8h shift)	2,000	4,000
Hospital Navigator (per day)	1,500	2,500
GP Online / Home Visit	500 / 1,500	1,500 / 3,000
Specialist Online / Premium Home Visit	1,000 / 5,000	3,000 / 12,000

Commission: 10% on nurse/doctor/lab/equipment/B2B bookings, 8% on medicine delivery. Surcharges: +20% peak (10pm-6am), +50% emergency. Launch offers: 5% commission for first 3 months, PKR 200 first-booking discount, PKR 100 digital-payment cashback.

3.7 Payments, Wallet & Payouts

Methods: **JazzCash, Easypaisa, Debit/Credit Card, Bank Transfer, Cash on Delivery** — all optional in the UI as pluggable payment methods (see §6.6, manual system for now). Wallet with cashback: PKR 200 first top-up, 2% per booking, PKR 100 cash→digital switch, PKR 300 referral, PKR 500 at 10 bookings, 3% on top-ups >PKR 10,000. Professional payout schedule: daily / weekly / monthly / on-demand, min payout PKR 500.

3.8 Ratings, Reviews & Notifications

5-category star rating (Punctuality, Professionalism, Medical Skills, Communication, Overall). Anonymous option (“Verified Patient” — never anonymous to Homigo Care internally). Every review AND every professional reply is **admin-moderated** before publishing. Two-way rating (professionals rate patients too). Review cashback: PKR 50 / 75 (50+ words) / 300 (10 reviews). Notifications: 3 priority tiers (mandatory/important/optional), Urdu+English, 30-day history, quiet hours (Priority-1 always breaks through).

3.9 Referral System

Unique code per patient, PKR 300 cashback both sides on first completed booking, live monthly leaderboard, PKR 10,000 top-referrer prize, milestone bonuses at 5/10/25 referrals.

3.10 Settings Decisions (carried over as-is)

Light mode only (no dark mode) app-wide for v1. Saved payment methods optional. Account deletion: 30-day soft-delete window with reminders, then permanent erase (transaction records anonymized and retained for legal compliance only).

4. Platform Strategy — Three Separate Builds

Per your explicit instruction, Homigo Care ships as **three independent front-ends** sharing one backend — **not** one React Native codebase pretending to be both:

1. **Android App** — a genuine native Android app (official Google Play build quality)
2. **iOS App** — a genuine native iOS app, built to feel **more premium** than the Android app (per your instruction), independent codebase
3. **Web App** — one responsive codebase that renders **three distinct experiences**: Desktop, Android-mobile-browser, iOS-mobile-browser

4.1 Android Native App

Layer	Technology	Why
Language	Kotlin	Google’s official, modern Android language
UI Toolkit	Jetpack Compose + Material 3 (Material You)	Google’s current official declarative UI system — this is what makes an app feel

Architecture	MVVM + Clean Architecture (Repository pattern)	“real Android,” not a wrapped web app Testable, scalable, standard for production apps
Async	Kotlin Coroutines + Flow	Native async, works perfectly with Compose state
Navigation	Jetpack Navigation Compose	Official navigation graph
DI	Hilt	Official Google DI library
Local cache	Room (SQLite)	Offline booking history, cached Health Passport
Networking	Retrofit + OkHttp	Talks to Firebase/Supabase REST & Cloud Functions
Realtime	Firebase Firestore SDK (Android) + Firebase Cloud Messaging	Live tracking, booking status, push notifications
Maps	Google Maps SDK for Android (free tier) or OSMDroid (fully free, OpenStreetMap)	See \$6.5
Build/Distribute	Gradle → signed .aab/.apk via GitHub Actions	Source lives on GitHub; a GitHub Actions workflow runs <code>./gradlew bundleRelease</code> , signs it, and uploads the artifact — this is your “GitHub → converts to Android app” pipeline

Why not React Native (deviation from the old Developer Guide): the old guide recommended React Native because it targeted *one* codebase for both platforms. You now want two independently premium, officially-native apps — Kotlin/Compose for Android and Swift/SwiftUI for iOS deliver the authentic native look, native performance, and access to each platform’s real icon/asset system (Material Symbols, SF Symbols) that a cross-platform framework cannot fully replicate.

4.2 iOS Native App (Premium Tier)

Layer	Technology	Why
Language	Swift	Apple’s official language
UI Toolkit	SwiftUI	Apple’s current official declarative UI, matches iOS 17/18 design language natively (rounded continuous corners, native blur, native haptics)
Architecture	MVVM + Combine / Swift Concurrency (async/await)	Standard modern iOS architecture
Local cache	SwiftData (or Core Data)	Offline cache, matches Room’s role on Android
Networking	URLSession / Alamofire	Talks to the same Firebase/Supabase backend
Realtime	Firebase Firestore SDK (iOS) + Firebase Cloud Messaging (APNs bridge)	Same real-time backend as Android — one shared backend, two native clients
Maps	MapKit (native, free, no key needed) or Google Maps SDK for iOS	MapKit is the “premium native” choice — feels 100% like a first-party iOS app Important honest caveat: Apple requires a paid Apple Developer Program membership (US \$99/year) to install on real devices via TestFlight or to publish to the App Store. This is the one cost that cannot be made free — everything else in this blueprint (backend, hosting, storage, email) is free. Budget for this when you’re ready to test on real iPhones or launch.
Build/Distribute	Xcode Cloud or GitHub Actions macOS runner → signed .ipa	

Why iOS gets the more premium visual treatment: SwiftUI’s native materials (blur, depth, spring animations), SF Symbols’ multicolor/hierarchical rendering, and Apple’s Human Interface Guidelines produce a naturally more “premium” feel with less custom work — lean into large-title navigation bars, native sheet presentations, SF Symbols in every icon slot, and haptic feedback (`UIImpactFeedbackGenerator`) on every primary action (booking confirm, SOS hold, PIN success).

4.3 Web App — One Codebase, Three Responsive Personalities

Layer	Technology	Why
Framework	Next.js 14+ (App Router) + React + TypeScript	Industry standard, deploys natively to Vercel, supports both the public site and can share components with the Admin Panel
Styling	Tailwind CSS	Fast, consistent design-token driven styling — maps 1:1 to the color/spacing tokens in §5
State	Zustand or React Context	Lightweight, no backend lock-in

Realtime	Firebase JS SDK (Firestore + Auth)	Same backend as the native apps
Hosting	Vercel (free Hobby tier)	Auto-deploys from GitHub on every push, free SSL, free custom domain support, serverless functions for anything the client shouldn't do directly (e.g. admin actions)

Serving three layouts from one deployment — implementation approach: 1. Detect device class on the server (middleware, via user-agent) **and** re-verify on the client with a CSS media-query/viewport check (never trust user-agent alone). 2. Maintain **three layout shells** behind one router: DesktopShell, MobileAndroidShell, MobileIOSShell. Screens/pages import shared business-logic hooks but render inside whichever shell is active. 3. **Desktop Shell** — full sidebar navigation, multi-column dashboards, hover states, wide booking tables — built for mouse + keyboard. 4. **Mobile-Android-Web Shell** — bottom navigation bar using **Material 3** styling (elevated FAB for Emergency, ripple effects, Roboto-style spacing) — feels like a Material web app. 5. **Mobile-iOS-Web Shell** — bottom tab bar using **Apple HIG** styling (SF Pro-style font stack via -apple-system, translucent blurred tab bar via backdrop-filter, iOS-style large titles) — feels like using Safari on an iPhone with an iOS-native app. 6. All three shells pull from the **same design tokens** (§5) so brand colors/logo never drift — only chrome, spacing, and navigation metaphors change per shell.

5. Design System (unchanged brand tokens — apply identically on all 3 platforms)

Token	Hex	Usage
Primary Teal	#1B6B63	Buttons, headers, active nav, logo “HOMIGO”
Dark Teal	#0D4A44	Splash gradient bottom, hover/pressed states
Header Deep Emerald	#0b5545	Premium header bars (per HOMIGO PROMPTS reference doc)
Secondary Orange	#F5823A	Logo “CARE”, accents, cashback highlights, voice-search mic
White	#FFFFFF	Backgrounds, cards
Light Gray	#F5F5F5	Input field backgrounds
Dark Gray	#333333	Body text
Medium Gray	#888888	Placeholder / secondary text
Error Red	#E53935	SOS button, error states
Success Green	#43A047	Confirmed/online status

Geometry: cards 15px corner radius + subtle flat shadow; buttons & input fields 25px corner radius (fully rounded pill style); Emergency button always red with a pulsing animation. **Typography:** Sans-serif throughout (Android → **Roboto/Google Sans**; iOS → **SF Pro**; Web → **Plus Jakarta Sans** with Noto Nastaliq Urdu for Urdu text) — H1 22px Bold, H2 18px SemiBold, Body 14px, Small 12px, Button 16px Bold. **Mode:** Light mode only across all three platforms for v1.

5.1 Mobile-Safe Layout Rules (mandatory on every screen, every platform)

These rules exist so **no element is ever cut off, hidden behind a notch/status bar, or unreachable** on any real device — this was an explicit requirement:

- Safe-area first.** Android: respect WindowInsets (status bar, gesture nav bar, cutouts) using Scaffold’s built-in insets in Compose. iOS: use .safeAreaInset() / GeometryReader — never hardcode top/bottom padding. Web: env(safe-area-inset-top), env(safe-area-inset-bottom) plus viewport-fit=cover.
- No fixed pixel heights for content that can grow** (Urdu text, long names, long addresses) — always measure with intrinsic/flexible sizing (wrap_content/IntrinsicSize/auto), never overflow: hidden on primary content.
- Minimum tap target 44×44 pt (iOS) / 48×48 dp (Android)** on every button, icon, and list row — this directly satisfies “clickable and adjustable” comfortably.
- Test at 3 breakpoints minimum per platform:** small phone (360×640 / iPhone SE), standard phone (390×844 / Pixel), and large phone (430×932 / Pro Max / large Android) — every screen must reflow, not just scale down and clip.
- Bottom navigation and the SOS/Emergency button are always exempt from scroll** — pin them above the safe-area inset, never let page content scroll underneath and cover them permanently.
- RTL readiness:** because Urdu is a first-class language, every layout uses direction-aware spacing (start/end, not hardcoded left/right) so the Urdu toggle doesn’t break alignment.
- Loading/error states never leave blank space** — every async screen shows a skeleton or spinner, and every empty list shows a real empty-state illustration + short text (ties into the zero-fake-data rule in §9).

6. Icons & Visual Assets Strategy

Your instruction was explicit: **no AI-emoji-style icons**, use **real, official platform icon systems** wherever one exists, and only fall back to AI generation for illustrations/PNG art that has no official source — with prompts batched **20-50 images per request**, in **JSON**, one batch per theme (never one prompt per single image).

6.1 Icon Libraries — Use the Real, Official, Free Ones

Platform	Official Icon System	How to include it	License
Android	Material Symbols (Google’s official icon library — thousands of outlined/filled/rounded glyphs)	Add the <code>androidx.compose.material.icons.extended</code> artifact, or bundle Material Symbols variable font	Apache 2.0 — free for commercial use
	SF Symbols (Apple’s official system icon font, 6,000+ glyphs, built into every iOS SDK)	<code>Image(systemName: "cross.case.fill")</code> etc. — zero download needed, ships with Xcode	Free, but restricted to use within apps built for Apple platforms (do not export SF Symbols glyphs into the Android app or a public web page — use Material Symbols / Lucide there instead, styled to <i>look</i> consistent)
	Lucide Icons (already recommended in your original Developer Guide as “Lucide React Native” — same family, MIT license, free, open source)	lucide-react npm package	MIT — fully free
Web (all 3 shells)			

This gives every platform its **real native icon identity** (Android looks Android, iOS looks iOS) while Lucide keeps the web version visually consistent with both without violating Apple’s SF Symbols usage terms.

6.2 Illustrations & PNG Assets — Two Legitimate Paths

Path A — Curated free libraries (fastest, zero AI needed): - **unDraw** (undraw.co) — free SVG illustrations, recolorable to exact brand teal #1B6B63 - **Storyset** (storyset.com) — free animated/static illustrations, professional look - Both already recommended in your original Developer Guide for onboarding art — reuse them for empty states, success states, and error states too.

Path B — AI-generated assets, for anything Path A can’t cover (custom badges, Pakistan-context illustrations, role-specific avatars, achievement icons): use the **batched JSON prompt system** below. You paste each batch into your AI image generator (Gemini/Imagen, Midjourney, etc.), generate the set, then drop the resulting PNGs into the app’s asset folders at the paths indicated.

6.3 Batched AI Image-Generation Prompt System

Rules for every batch: 20–30 images per JSON request (never 1-image-per-prompt), every item specifies subject, exact background treatment (transparent PNG vs. solid brand color), art style, and target asset path, so nothing needs a follow-up clarifying prompt.

```
{
  "batch_id": "HGC-ASSET-BATCH-01",
  "batch_name": "Onboarding & Empty-State Illustrations",
  "global_style": {
    "art_style": "flat 2D vector illustration, clean line art, no gradients except brand teal-to-dark-teal where noted",
    "palette": ["#1B6B63", "#0D4A44", "#F5823A", "#FFFFFF", "#F5F5F5"],
    "background": "transparent PNG (alpha channel), unless an item says otherwise",
    "export": "PNG, 2048x2048, transparent background, centered subject with 10% padding",
    "forbidden": ["photorealism", "3D render", "watermarks", "text baked into image", "real human faces/photos", "any copyrig
character or logo"]
  },
  "images": [
    {"id": "onb_01", "path": "/assets/illustrations/onboarding_home_care.png", "prompt": "Flat vector illustration of a frien
nurse in teal scrubs kneeling beside an elderly patient seated in a home living room, warm and reassuring mood, minimal line art,
teal and orange accent colors only, transparent background"},
    {"id": "onb_02", "path": "/assets/illustrations/onboarding_verified.png", "prompt": "Flat vector illustration of a large
verified checkmark shield hovering above a stack of documents and a professional ID card, teal shield with white checkmark, orange
accent sparkle, transparent background"},
    {"id": "onb_03", "path": "/assets/illustrations/onboarding_safety.png", "prompt": "Flat vector illustration of a GPS map
inside a protective shield outline, small radiating safety rings around it, teal and orange color scheme only, transparent
background"},
    {"id": "onb_04", "path": "/assets/illustrations/empty_bookings.png", "prompt": "Flat vector illustration of an empty cale
page with a small clock icon, friendly and light, gray and teal linework, transparent background, no text"},
    {"id": "onb_05", "path": "/assets/illustrations/empty_notifications.png", "prompt": "Flat vector illustration of a bell i
with a small sleeping crescent moon beside it, soft gray and teal linework, transparent background, no text"},
    {"id": "onb_06", "path": "/assets/illustrations/empty_reviews.png", "prompt": "Flat vector illustration of a single outli
star with a small speech bubble, gray and orange linework, transparent background, no text"},
    {"id": "onb_07", "path": "/assets/illustrations/empty_wallet.png", "prompt": "Flat vector illustration of a simple wallet
with a small coin floating above it, teal and orange linework, transparent background, no text"},
    {"id": "onb_08", "path": "/assets/illustrations/success_booking_confirmed.png", "prompt": "Flat vector illustration of a
circular teal checkmark badge with small celebratory confetti dots in orange, transparent background"},
    {"id": "onb_09", "path": "/assets/illustrations/error_connection.png", "prompt": "Flat vector illustration of a friendly
icon with a small disconnected plug symbol beside it, gray and teal linework, transparent background, no text"},
    {"id": "onb_10", "path": "/assets/illustrations/health_passport_hero.png", "prompt": "Flat vector illustration of a close
digital passport booklet with a small padlock on the cover and a heartbeat line motif, teal cover, orange heartbeat line,
transparent background"},
    {"id": "onb_11", "path": "/assets/illustrations/family_account.png", "prompt": "Flat vector illustration of three simple
```

```
silhouette figures of varying heights (adult, child, elderly) standing together holding hands, minimal line art, teal silhouettes
with orange accent hearts, transparent background"},
  {"id": "onb_12", "path": "/assets/illustrations/lab_test.png", "prompt": "Flat vector illustration of a test tube rack wi
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orange only, transparent background"},
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line icon centered, flat vector style, transparent background outside the circle"},
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medicine capsule line icon centered, flat vector style, transparent background outside the circle"},
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with a simple white compass line icon centered, flat vector style, transparent background outside the circle"},
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blood-drop line icon centered, flat vector style, transparent background outside the circle"},
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x-ray scan line icon centered, flat vector style, transparent background outside the circle"},
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]
}
```

```
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    "export": "PNG, 1024x1024, transparent background",
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    {"id": "badge_verified_pharmacy", "path": "/assets/badges/verified_pharmacy_partner.png", "prompt": "Flat circular teal badge medallion with a white medicine capsule icon in the center, no text, transparent background"},
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    {"id": "badge_icu_specialist", "path": "/assets/badges/specialization_icu.png", "prompt": "Flat circular orange badge medallion with a white heart-monitor line icon in the center, no text, transparent background"},
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    {"id": "badge_10_bookings", "path": "/assets/badges/ten_bookings_milestone.png", "prompt": "Flat circular teal badge medallion with a white number ten stylized inside a checkmark shape, no text besides the numeral, transparent background"},
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  ]
}
```

Two more batches to generate the same way when needed (not fully expanded here to keep this document usable — follow the exact JSON structure above, 20–30 items each): - HGC-ASSET-BATCH-04 — **Professional role avatars** (flat generic silhouette avatars per role: nurse, GP, specialist, lab technician, pharmacist, equipment technician, hospital navigator — used as default profile pictures before a real photo is uploaded; never depict a real person). - HGC-ASSET-BATCH-05 — **Admin panel dashboard iconography** (finance, disputes, verification queue, live map, communications, analytics — flat line icons matching Lucide’s style for visual consistency with the icons already used elsewhere).

7. Backend Architecture — 100% Free-Tier Stack

Your platform list is fully supported. Here is exactly **what each service is responsible for**, so nothing overlaps or gets duplicated by accident.

7.1 Division of Responsibility

Service	Responsible For	Free Tier Notes
Firestore	Sign up/login: Email+Password, “Continue with Google”, Guest/Anonymous browsing (per your list)	Free (Spark plan) up to very high monthly active user counts
Realtime Database	Real-time data that changes constantly: live GPS tracking during a visit, booking status updates, chat/notes, online/offline professional status, notification feed.	Free Spark plan: 1 GiB storage, 50K reads/20K writes per day — plenty for MVP/testing
Firestore	Collections auto-create on first write — no manual schema step needed.	

Firebase Cloud Messaging (FCM)	Push notifications on Android, iOS (via APNs bridge), and Web	Completely free, no limit
Supabase (Postgres)	Structured, relational, reportable data: user profiles, professional documents & verification status, bookings ledger, payments/commission ledger, reviews, admin tables, role-requirements config table. Postgres is the right tool here because bookings connect to a patient + professional + address + payment + review all at once — exactly the relational need identified in your original Developer Guide, now on a free Postgres instead of paid AWS RDS. Tables can be auto-created from a migration script you run once (supabase db push), after which the app just reads/writes rows — no manual dashboard clicking per table.	Free tier: 500 MB database, 5 GB bandwidth/month, unlimited API requests
Supabase Row Level Security (RLS)	Enforces “a patient can only read their own bookings,” “a professional can’t see other professionals’ payouts,” etc. — directly at the database level	Included free
Cloudinary	Every image/file: CNIC front/back, profile photos, degree/license scans, prescription photos, lab report uploads, receipt photos, onboarding illustrations, icon PNGs generated in \$6	Free tier: 25 GB storage + 25 GB monthly bandwidth — signed/private upload presets keep documents non-public (see §8)
Vercel	Hosts the Web App (Next.js, all 3 responsive shells) and the Admin Panel (see Document 2), auto-deploys on every git push, serverless API routes for anything requiring a secret key (Cloudinary signing, Supabase service-role calls)	Free Hobby tier covers a growing app comfortably
EmailJS	Transactional email without running your own mail server: booking confirmation email, password-reset email, document-approved/rejected email, weekly earnings summary for professionals, admin-to-user replies	Free tier: 200 emails/month — enough for early testing; upgrade only once volume demands it
GitHub	Single source of truth for all three codebases (homigo-android, homigo-ios, homigo-web) in one organization; GitHub Actions builds the signed Android .aab, and (with a Mac runner) the iOS .ipa; Vercel listens to the homigo-web repo for auto-deploy	Free for public/private repos + free Actions minutes tier

7.2 What Stays “Optional / Plug In Later” (per your instruction)

Feature	Status Now	How It’s Kept Ready
SMS OTP (phone verification)	Skipped for launch — use Email+Password / Google / Guest instead	Build the AuthProvider interface with an OtpProvider stub already wired into the sign-up flow’s UI (the phone-number field and “Verify Code” screen ship in the design, just backed by a “coming soon” flag) so switching on Twilio/Firebase Phone Auth later is a config change, not a rebuild
Payment Gateway (JazzCash/Easypaisa/Card API)	Manual system (see §7.3)	Payment method enum already includes all types; only the “auto-confirm” step is manual today
Location Services	Free tier now	See below

7.3 Payments — Manual System (as instructed)

No live payment gateway integration yet. Flow: 1. Admin Panel → **Payment Accounts** module: Super/Finance Admin adds any number of receiving accounts (JazzCash number, Easypaisa number, bank IBAN+title), each with a QR/screenshot if useful. 2. At checkout, the patient sees the active receiving accounts for their chosen method, sends the money outside the app, then enters the **transaction ID / reference number** (and optionally uploads a screenshot to Cloudinary) in the booking summary screen. 3. Booking status becomes Payment Pending Verification. Finance Admin sees it in the Admin Panel’s finance queue, confirms it manually against the real bank/wallet statement, and marks it **paid** — this releases the booking to the professional. 4. Cash on Delivery needs no verification step; it’s marked paid at visit completion by the professional. 5. This same manual pattern extends to **professional payouts** — Finance Admin manually transfers to the professional’s JazzCash/Easypaisa/bank

and marks the payout Completed in the dashboard (see Document 2, §4).

This is intentionally simple so you can test the entire booking economy for real, with real money movements outside the app, before wiring an automated gateway.

7.4 Location & Maps — Free Tier

Two supported options, pick one at launch: - **Option A (fully free forever): OpenStreetMap** tiles + **Leaflet** (web) / **OSMDroid** (Android) / **MapKit is free on iOS already**, plus **Nominatim** for free address search/geocoding. Zero cost at any scale, self-hostable if Nominatim’s public rate limit is ever a concern. - **Option B (richer UX, free up to a threshold): Google Maps Platform** — Google provides a recurring monthly free usage credit that comfortably covers a small-to-mid size app’s Maps/Geocoding/Places calls; upgrade path is simple since it’s the same SDK your original Developer Guide already specified. - **Recommendation:** start with Option A for Android + Web (100% free, no credit card needed to start) and use native **MapKit** for iOS (also 100% free); revisit Google Maps only if you need richer place-search/traffic data later.

7.5 Auto-Created Collections & Tables (adapted from the original DB schema)

Firestore collections (auto-create on first write): bookings_live (active tracking state), presence (professional online/offline + last GPS ping), chat_threads, notifications_feed, booking_status_events.

Supabase Postgres tables (create once via migration script):

Table	Stores
users	All accounts — id, name, phone/email, role, auth_uid (links to Firebase Auth)
patients	CNIC (encrypted), emergency contact, blood group, patient_type (adult/minor/elderly)
professionals	role_id (FK → role_requirements), hospital affiliation, verification_status, badges
role_requirements	Dynamic config: required documents/tests per role (§2.2)
addresses	Saved patient addresses
bookings	patient_id, professional_id, service, date, status, price, booking_type
health_passport	Medical history, conditions, allergies, medications
vitals	BP, sugar, weight, temperature time-series
documents	Cloudinary URL/public_id, doc type, owner, expiry date
reviews	5-category scores, text, anonymous flag, moderation_status
payments	booking_id, method, amount, status (pending_verification/paid/refunded), reference_no
payment_accounts	Admin-managed receiving accounts (JazzCash/Easypaisa/bank)
wallet_transactions	Wallet credits/debits per patient
assessment_tests	Attempt scores per phase per professional
supply_requests	Nurse-requested supply items, approval status
cms_content	All editable site text — About Us, Support contact info, FAQ, legal pages (see Document 2 §3) — the app/web/admin all read from here, nothing is hardcoded
access_logs	Every Health Passport access — who, when, which booking

8. Security (adapted to the free-tier stack)

Layer	Protection
Login	Firebase Auth (Email/Password + Google OAuth + Anonymous Guest) issues a secure ID token; Supabase verifies that token via a Firebase→Supabase JWT bridge (custom claims) before allowing any database read/write
Passwords	Never touched by your code — Firebase Auth stores and hashes them internally
Data in transit	HTTPS/TLS everywhere by default (Firebase, Supabase, Cloudinary, Vercel all enforce TLS)
Data at rest	CNIC numbers and other sensitive columns encrypted at the application layer before insert (e.g. pgcrypto in Supabase) in addition to Postgres’s own at-rest encryption
Row-level access	Supabase Row Level Security policies: a patient row is only readable/writable by that patient’s auth id; a professional cannot query another professional’s payouts; only admin role JWT claims can read payment_accounts/finance tables
	Cloudinary private/authenticated delivery — files are never publicly reachable by a guessed URL; the app requests a

Documents (CNIC, prescriptions, licenses)	short-lived signed URL from a Vercel serverless function each time a document needs to be viewed
Health Passport	Unchanged patient-controlled PIN model (§3.3) sits on top of normal auth — even a logged-in professional cannot read Health Passport fields without the per-visit PIN; every read is written to <code>access_logs</code>
API / serverless functions	Rate limiting on Vercel Edge Middleware; input validation (Zod schemas) on every request body; CORS restricted to the app’s own domains
Admin Panel	Separate Supabase auth role (<code>admin</code>), 2FA recommended (Firebase Auth supports TOTP), and IP-based access notes recommended once you have a fixed office IP (VPN-only was the original recommendation; with Vercel hosting, the practical equivalent is <code>admin-role JWT + 2FA + login alert emails via EmailJS</code>)

9. Zero Fake Data — Real-Base System Policy

This is a hard rule across Android, iOS, Web, and the Admin Panel:

1. **No seed scripts that insert demo users, demo bookings, demo reviews, or demo professionals.** The database starts genuinely empty (aside from the `role_requirements` config table, which is configuration, not user data).
2. **Every dashboard stat, chart, and list must have a real empty state** — e.g. Admin “Today’s Revenue: PKR 0” with a message like “No bookings yet — figures will appear as real bookings come in,” not a hidden fake number.
3. **The first data in the system is created by you, for real, through the actual sign-up/booking flow** — register yourself as a patient, register a real (even if it’s you again) professional profile, approve it from the Admin Panel, then place a real test booking. This is your live QA method, not fake data.
4. **AI-generated icons/illustrations (§6) are exempt** from this rule — they are UI chrome, not user/business data, so they’re fine to pre-load.
5. Any developer or AI coding agent building this must **never** add a `// TODO: remove this dummy data later` block — build every list/screen against the real (currently empty) data source from day one.

10. Screen Inventory Reference

Full screen-by-screen **layouts, copy, and behavior** for every screen below are already fully specified in your existing `HomigoCare_Complete_Developer_Document.docx` and `HomigoCare_Developer_Guide.docx` — **keep using those two documents as the pixel-level spec**; this blueprint’s job is to tell the AI builder *which platform/tech to build them in* and *how the icons/assets/backend plug in*. The table below is the master index so nothing gets skipped across all three platforms.

10.1 Patient App/Site — 32 Screens

P-01 Splash · P-02–04 Onboarding (3 screens) · P-05 Login · P-06 Sign Up · P-07 OTP/Verify* · P-08–10 Profile Setup (Personal/Address/Emergency Contact) · P-11 Home Dashboard · P-12 Service Category · P-13 Browse Professionals · P-14 Professional Profile · P-15 Booking Details · P-16 Health Passport PIN · P-17 Supply Pre-Request · P-18 Booking Summary & Payment · P-19 Booking Confirmed · P-20 Live Tracking · P-21 Visit In Progress · P-22 Visit Complete · P-23 Rating & Review · P-24 Health Passport (5 tabs) · P-25 Emergency Screen · P-26 Lab Test Booking · P-27 Medicine Order · P-28 Wallet · P-29 Referral & Leaderboard · P-30 Notifications · P-31 Profile & Settings · P-32 Booking History.

*P-07 ships as an Email-verification-link / “Verify your email” step for launch since phone-OTP is deferred (§7.2); keep the OTP-style 6-box UI, just point it at an email code instead of SMS until Twilio/Phone-Auth is switched on.

Additional registration branch screens (from the Registration Guide, insert after OTP where relevant): “Who is this for?” selector (Myself / My Child / Elderly Family Member), Guardian-linked flow (PT-B), Family-assisted elderly flow (PT-C).

10.2 Professional App/Site — 15 Screens

PR-01 Splash · PR-02 Login · PR-03 Sign Up · PR-04 Document Upload (**dynamic, driven by role_requirements**) · PR-05 Verification Pending · PR-06–08 Assessment Test (Intro/Test/Results) · PR-09 Profile Activation · PR-10 Home Dashboard · PR-11 Incoming Booking Request (10-min countdown) · PR-12 Active Booking · PR-13 Visit Notes (mandatory) · PR-14 Earnings Dashboard · PR-15 Professional Profile Management.

10.3 Admin Panel — see Document 2 for the full, dedicated specification.

11. Recommended AI Coding Assistant & Model

You mentioned you have access to **Antigravity** (Google’s agentic coding IDE, built around the Gemini model family) — this is a solid fit for building three codebases against one shared backend, since it can hold long project context and run multi-file agent tasks.

Recommendation (check Antigravity’s current model picker before you start, since Google updates this list often):
- For **architecture, database schema, and full-screen generation work** (the heavy lifting — e.g. “build the Health

Passport module end-to-end”): use the **highest-reasoning “Pro” tier Gemini model available in Antigravity at the time** (the flagship “Pro/High-thinking” option). This is worth the extra latency for anything security-sensitive (auth, RLS policies, payment status logic) or that touches many files at once. - For **fast iteration** (small UI tweaks, one-screen changes, icon wiring, copy edits): use the **“Flash” tier** model in the same picker — it is dramatically faster/cheaper and is the tier Google specifically tuned for tight agent loops. - If Antigravity’s picker also offers non-Gemini options (it sometimes bundles alternatives), you can use those for a second opinion on tricky bugs, but keep Gemini as your default so context/memory stays consistent across the whole multi-repo project. - Practical workflow: feed Antigravity **this document + Document 2 + the original HomigoCare_Complete_Developer_Document.docx** as project context/“Knowledge,” then work **repo by repo** (homigo-android → homigo-ios → homigo-web → homigo-admin) rather than asking for all four at once — this keeps each agent run focused and reviewable.

12. Build Roadmap (phases carried over from the original spec, re-mapped to the new stack)

Phase 1 — MVP (build first, one platform at a time or in parallel if you have 2+ builders): Splash → Onboarding → Login/Sign Up (Email+Google+Guest) → Profile Setup (3 steps) → Home Dashboard → Service Category → Browse Professionals → Professional Profile → Booking Details → Booking Summary & Payment (manual) → Booking Confirmed. Professional side: Sign Up → Document Upload (dynamic) → Verification Pending → 3-Phase Assessment → Dashboard → Accept/Reject Booking. Backend: Firebase Auth + Firestore skeleton + Supabase core tables + Cloudinary upload presets + Vercel web deploy.

Phase 2 — Core Features: Live Tracking → Visit In Progress (both apps) → Visit Complete → Rating & Review → Health Passport (all 5 tabs) + PIN sharing → Supply Pre-Request → Notifications → Booking History → Professional Navigation/Earnings screens → GPS Safe Zone + SOS system → Admin Panel v1 (verification queue, bookings, emergency alerts — see Document 2).

Phase 3 — Full Feature Set: Emergency Screen (full), Lab Test Booking, Medicine Order, Equipment Rental, B2B Supply Portal, Wallet + Referral Leaderboard, full Admin Panel (finance, disputes, analytics, communications, CMS content management), Urdu voice search + icon-based booking + full Urdu toggle, combined bookings, recurring bookings.

Phase 4 — Post-Launch Roadmap (explicitly “Coming Soon” in the source docs, unchanged): Homigo Care Supply Store, auto-refill medicines, medicine interaction checker, barcode authenticity scanner, AI fake-review detection, IoT equipment monitoring, WhatsApp notifications, automated document verification, predictive analytics, regional languages (Punjabi/Pashto/Sindhi), video reviews, corporate accounts, insurance integration, buy-now-pay-later, phone OTP switch-on, live payment gateway switch-on, and the **Hospital Management System + Universal QR Patient Record** feature (full technical plan already exists in HomigoCare_HMS_QR_Feature_Urdu.pdf — it reuses the existing Health Passport PIN/consent model, just extends it to hospital terminals via a closed network first, then HL7 FHIR for open interoperability later).